

BETS

Advanced IP Subnetting Assignment

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Assignment: Advanced IP Subnetting Scenarios

Instructions: Read each scenario carefully. For each situation, calculate the new subnet mask (in both CIDR and decimal formats), the number of usable hosts per subnet, and the specific network ranges requested. Show your work.

Part 1: Class C Subnetting Scenarios

Scenario 1: The Small Law Firm

Network Block: 192.168.15.0/24 | Requirement: 4 subnets (Partners, Associates, Paralegals, Guest Wi-Fi)

Student Tasks

A) New subnet mask to create exactly 4 subnets:

To create 4 subnets, we need to borrow 2 bits from the host portion ($2^2 = 4$ subnets).

CIDR Notation	/26
Decimal Mask	255.255.255.192

B) Usable host IP addresses per subnet:

With /26, host bits = 6. Usable hosts = $2^6 - 2 = 64 - 2 = 62$

Usable Hosts/Subnet	62 hosts
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C) 2nd Subnet (Associates) Network ID, Usable Range, Broadcast:

Block size = $256 - 192 = 64$. Subnets increment by 64.

Network ID	192.168.15.64
Usable IP Range	192.168.15.65 – 192.168.15.126
Broadcast Address	192.168.15.127

Scenario 2: The Medical Clinic

Network Block: 200.50.25.0/24 | Requirement: Minimum 6 subnets

Student Tasks

A) Bits to borrow for at least 6 subnets, and total subnets created:

$2^2 = 4$ (not enough), $2^3 = 8$ (sufficient). Borrow 3 bits → 8 total subnets.

Bits Borrowed	3 bits
Total Subnets	8 subnets ($2^3 = 8$)

B) New subnet mask:

Original /24 + 3 borrowed bits = /27

CIDR Notation	/27
Decimal Mask	255.255.255.224

C) 1st Subnet (Reception) Network ID and Broadcast Address:

Block size = $256 - 224 = 32$. First subnet starts at .0.

Network ID	200.50.25.0
Broadcast Address	200.50.25.31

Scenario 3: The High School Computer Labs

Network Block: 192.168.100.0/24 | Requirement: 4 subnets, each supporting at least 50 hosts

Student Tasks

A) Subnet mask for 4 subnets supporting at least 50 usable hosts:

Need 4 subnets AND at least 50 hosts each. With /26: 2 bits borrowed = 4 subnets, host bits = 6, usable = 62. This satisfies both requirements.

CIDR Notation	/26
Decimal Mask	255.255.255.192

B) Maximum usable hosts this mask provides per subnet:

Max Usable Hosts	62 hosts per subnet ($2^6 - 2 = 62$)
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C) Usable IP Range for the 4th subnet:

Block size = 64. Subnets: .0, .64, .128, .192

4th Subnet Network	192.168.100.192
Usable IP Range	192.168.100.193 – 192.168.100.254
Broadcast	192.168.100.255

Scenario 4: The Coffee Shop Chain

Network Block: 198.100.50.0/24 | Requirement: Maximum subnets with at least 14 usable hosts each

Student Tasks

A) Most efficient subnet mask:

Need at least 14 usable hosts: $2^4 - 2 = 14$ ✓ (4 host bits). Borrow 4 bits → /28.

CIDR Notation	/28
Decimal Mask	255.255.255.240

B) Total subnets created:

Total Subnets	16 subnets ($2^4 = 16$)
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C) Network ID for the 3rd subnet:

Block size = 256 - 240 = 16. Subnets: .0, .16, .32, ...

3rd Subnet Network ID	198.100.50.32
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Scenario 5: The E-Commerce Startup

Network Block: 203.0.113.0/24 | Requirement: 12 distinct subnets for development squads

Student Tasks

A) Bits borrowed for at least 12 subnets:

$2^3 = 8$ (not enough), $2^4 = 16$ (sufficient). Borrow 4 bits.

Bits Borrowed	4 bits ($2^4 = 16$ subnets)
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B) Usable IP addresses per development squad:

Host bits remaining = 8 - 4 = 4. Usable = $2^4 - 2 = 14$

Usable Hosts/Subnet	14 hosts
Subnet Mask (CIDR)	/28
Subnet Mask (Decimal)	255.255.255.240

C) Last (highest) available subnet — Network ID, Usable Range, Broadcast:

Block size = 16. 16th subnet starts at: $15 \times 16 = 240$.

Network ID	203.0.113.240
Usable IP Range	203.0.113.241 – 203.0.113.254
Broadcast Address	203.0.113.255

Part 2: Class B Subnetting Scenarios

Scenario 6: The Regional Hospital

Network Block: 172.16.0.0/16 | Requirement: 50 separate subnets

Student Tasks

A) Bits borrowed and new subnet mask for at least 50 subnets:

$2^5 = 32$ (not enough), $2^6 = 64$ (sufficient). Borrow 6 bits. New mask = /16 + 6 = /22.

Bits Borrowed	6 bits ($2^6 = 64$ subnets)
CIDR Notation	/22
Decimal Mask	255.255.252.0

B) Usable hosts per subnet:

Host bits = $32 - 22 = 10$. Usable = $2^{10} - 2 = 1022$

Usable Hosts/Subnet	1,022 hosts
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C) 2nd Subnet — Network ID and Broadcast Address:

Block size in 3rd octet = $256 - 252 = 4$. Subnets increase by 4 in 3rd octet.

Network ID	172.16.4.0
Broadcast Address	172.16.7.255

Scenario 7: The Corporate Headquarters

Network Block: 172.25.0.0/16 | Requirement: Each subnet supports exactly 500 usable hosts

Student Tasks

A) Host bits required for at least 500 usable hosts:

$2^8 - 2 = 254$ (not enough), $2^9 - 2 = 510$ (sufficient). Need 9 host bits.

Host Bits Required	9 bits ($2^9 - 2 = 510$ usable hosts)
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B) New subnet mask:

Network bits = $32 - 9 = 23$. New mask = /23.

CIDR Notation	/23
Decimal Mask	255.255.254.0

C) Block Size (magic number) in the 3rd octet:

The 3rd octet mask value = 254. Block size = $256 - 254 = 2$.

Block Size (3rd Octet)	2 (subnets: .0.0, .2.0, .4.0, .6.0, ...)
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Scenario 8: The Smart City Project

Network Block: 128.10.0.0/16 | Requirement: At least 100 subnets

Student Tasks

A) Subnet mask for at least 100 subnets:

$2^6 = 64$ (not enough), $2^7 = 128$ (sufficient). Borrow 7 bits. New mask = $/16 + 7 = /23$.

CIDR Notation	/23
Decimal Mask	255.255.254.0

B) Usable IP addresses per city grid subnet:

Host bits = $32 - 23 = 9$. Usable = $2^9 - 2 = 510$

Usable Hosts/Subnet	510 hosts
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C) Network ID for the 3rd subnet:

Block size = 2 in 3rd octet. Subnets: .0.0, .2.0, .4.0, ...

3rd Subnet Network ID	128.10.4.0
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Scenario 9: The University Wi-Fi Zones

Network Block: 172.31.0.0/16 | Requirement: Each subnet supports at least 2,000 wireless devices

Student Tasks

A) Subnet mask for at least 2,000 usable hosts:

$2^{10} - 2 = 1022$ (not enough), $2^{11} - 2 = 2046$ (sufficient). Need 11 host bits. Mask = $/21$.

CIDR Notation	/21
Decimal Mask	255.255.248.0

B) Total subnets this configuration allows:

Subnet bits borrowed = $21 - 16 = 5$. Total subnets = $2^5 = 32$

Total Subnets	32 subnets
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C) Usable IP Range for the 1st subnet:

Block size in 3rd octet = $256 - 248 = 8$. First subnet: 172.31.0.0

Network ID	172.31.0.0
Usable IP Range	172.31.0.1 – 172.31.7.254
Broadcast	172.31.7.255

Scenario 10: The Manufacturing Plant

Network Block: 150.20.0.0/16 | Requirement: Exactly 16 equal-sized subnets

Student Tasks

A) Subnet mask for exactly 16 subnets:

$2^4 = 16$. Borrow 4 bits. New mask = $/16 + 4 = /20$.

CIDR Notation	/20
Decimal Mask	255.255.240.0

B) Usable hosts per assembly line subnet:

Host bits = $32 - 20 = 12$. Usable = $2^{12} - 2 = 4094$

Usable Hosts/Subnet	4,094 hosts
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C) 5th Subnet — Network ID, Usable Range, Broadcast Address:

Block size = $256 - 240 = 16$ in 3rd octet. 5th subnet: $4 \times 16 = 64$ in 3rd octet.

Network ID	150.20.64.0
Usable IP Range	150.20.64.1 – 150.20.79.254
Broadcast Address	150.20.79.255